

AMOXIKLAV

Powder

(FOR VETERINARY USE ONLY)

COMPOSITION

Amoxicillin (trihydrate)	500 mg
Clavulanic acid (potassium)	125 mg
Exp q.s to	1 g

PRODUCT DESCRIPTION

The finished product is a white color powder.

TARGET SPECIES

Pigs

PHARMACODYNAMIC

The product is a combination of a β -lactam antibiotic with a β -lactamase inhibitor which restores the potency of amoxicillin against strains producing β -lactamases. Amoxicillin is a bactericidal antibiotic, which acts by inhibiting the synthesis of bacterial cell walls during bacterial multiplication. It inhibits cross-linkage between the linear peptidoglycan polymer chains in the cell wall of Gram-positive bacteria. Broad spectrum penicillin antibiotic amoxicillin is active also against a limited range of Gram negative bacteria where the outer layer of bacteria cell-wall is made up of lypopolysaccharide and protein. There are three principal mechanisms of resistance to β -lactam antibiotics: production of β -lactamases, alteration of the PBP, diminished outer membrane permeation. One of the most important is inactivation of penicillin antibiotic with β -lactame enzymes produced by certain bacteria. These enzymes cleave the penicillin β -lactam ring and render the penicillin drug inactive. Clavulanic acid acts as inhibitor of bacterial β -lactamases. It prevents destruction of the β -lactam ring and penicillins by β -lactamase enzymes. The reaction is irreversible and both the enzyme and the clavulanate are destroyed while the antibiotic activity is preserved. The role of the clavulanic acid in the combination is not only to act as a β -lactamases inhibitor. Clinical efficacy is dependent upon a number of factors including not only intrinsic antibacterial properties but also a positive interaction with host defenses. After exposure to an antibacterial compound, the resulting alteration of cell-wall integrity and changes in bacterial expression of surface proteins, surface charges and hydrophobicity can influence the rate of phagocytosis and the extent of intracellular killing of bacteria.

An effect on the rates of phagocytosis and the intracellular killing functions of polymorphonuclear leucocytes was demonstrated in experimental studies. Susceptibility and resistance patterns can vary with geographical area and bacterial strain, and may change over time.

PHARMACOKINETIC

The plasma pharmacokinetic properties of amoxicillin and clavulanic acid are relatively similar. Both compounds are stable in acid environment of the gastrointestinal tract. After oral administration, amoxicillin and clavulanic acid are readily absorbed. The absorption following oral administration does not appear to be inhibited by presence of food in the alimentary tract.

Both compounds penetrate well into diverse tissue fluids (pleural, synovial, peritoneal fluids) and inflammatory exudates but not through the blood-brain barrier. Both compounds are largely eliminated by renal excretion. Elimination half-lives of amoxicillin and clavulanic acid are not significantly different (i.e. 0.73 and 0.67h respectively). Repeated treatment does not appear to result in any accumulation of amoxicillin and clavulanic acid. Therapeutic concentrations of amoxicillin and clavulanic acid are achieved approximately one hour after dosing and may persist for several hours after administration.

INDICATIONS

Treatment of clinical infections

- Respiratory tract infections caused by *Actinobacillus pleuropneumoniae*, *Pasteurella multocida*, *Streptococcus suis*.
- Gastrointestinal infections caused by *Clostridium perfringens*, *E. coli* and *Salmonella Typhimurium*

Where the causative pathogens are beta-lactamase producing strains of bacteria susceptible to amoxicillin in combination with clavulanic acid and where clinical experience and/or susceptibility testing indicates the combination as the drug of choice.

WARNINGS AND PRECAUTIONS

1. Special warnings:

The product should be given to pigs only.

2. Special precautions for use:

i. Special precautions for use in animals:

The uptake of medication by animals can be altered as a consequence of illness. In case of insufficient uptake of feed/water animals should be treated parenterally. Use of the product should be based on susceptibility testing and should take into account official national and regional policies with

respect to the use of broad spectrum antibiotics. Do not use in cases of bacteria sensitive to narrow spectrum penicillins or to amoxicillin as a single substance. Use of the product deviating may increase the prevalence of bacteria resistant to amoxicillin and clavulanic acid, and may decrease the effectiveness of treatment with other β -lactam antibiotics, due to the potential for cross resistance. Due to the resistance rate detected in some countries in porcine isolates of *E. coli* against amoxicillin in combination with clavulanic acid the product should be used for the treatment of infections caused by *E. coli* only after susceptibility testing has been carried out. Administration of the product should not be used as a method to control non-clinical salmonella infections in pig herds. It is strictly recommended, that the product should not be used as a tool of salmonella control programmes. In the case of a history of MRSA on a farm it is inappropriate to use a combination of amoxicillin and clavulanic acid as there is a likelihood to co-select for MRSA. The use of the product should be combined with good management practices e.g. good hygiene, proper ventilation, no overstocking.

ii. Special precautions to be taken by the person: Administering the veterinary medicinal product to animals: Penicillins and cephalosporins may cause hypersensitivity (allergy) following injection, inhalation, ingestion or skin contact. Hypersensitivity to penicillins may lead to cross reactions to cephalosporins and vice versa. Allergic reactions to these substances may occasionally be serious. Do not handle this product if you know you are sensitised or if you have been advised not to work with such preparations. Handle this product with great care to avoid exposure, taking all recommended precautions.

If you develop symptoms following exposure such as a skin rash, you should seek medical advice and show the doctor this warning. Swelling of the face, lips or eyes or difficulty with breathing are more serious symptoms and require urgent medical attention. Avoid inhalation of dust. Wear either a disposable half-mask respirator conforming to European Standard EN149 or a non-disposable respirator to European Standard EN140 with a filter to EN143. Wear gloves during preparation and administration of medicated water. Wash any exposed skin after handling the product or medicated water. Wash hands after use.

- Keep medicine out of reach of children.
- Jauhkan ubat dari kanak-kanak

CONTRA-INDICATIONS

Do not use in case of hypersensitivity to penicillin or cephalosporin antibiotics or any of the excipients. The product should not be given to rabbits, guinea pigs, hamsters, gerbils or small herbivores.

ADMINISTRATION & DOSAGE

Administration in drinking water.

Give 10mg of amoxicillin (in the form of trihydrate) and 2.5mg of clavulanic acid (in the form of potassium salt) per kg of body weight twice daily, i.e. 2g of product per 100 kg body weight twice daily. Treat for 5 days. For the calculation of the amount to be administered every 12 hours the following formula can be used:

Number of pigs x average body weight (kg) x dose rate (0.02 g of product / kg body weight) twice daily.

During the twice daily treatment periods, medicated drinking water should be the only water supply available. After all the medicated drinking water has been consumed, resume the supply of unmedicated water. To ensure correct dosage body weight of animals should be determined as accurately as possible to avoid underdosing.

The intake of medicated drinking water depends on the clinical conditions of the animals as well as on the weather/temperature. In order to obtain the correct dosage the concentration of the product should be adjusted accordingly.

INTERACTION WITH OTHER MEDICAMENTS

In general penicillins may be inhibited by the antibiotics with bacteriostatic action such as macrolides, sulfonamides and tetracyclines. No specific data on interaction of the combination have been described in the veterinary literature available. Neomycin given orally inhibits the intestinal absorption of penicillin.

PREGNANCY AND LACTATION

Laboratory studies in rats and mice have not produced any evidence of mutagenicity, teratogenic and foetotoxic effects. The safety of the veterinary medicinal product has not been established during pregnancy and lactation. Use only accordingly to the benefit/risk assessment by the responsible veterinarian.

OVERDOSE AND TREATMENT

In case of severe hypersensitive reactions, the treatment should be discontinued and corticosteroids and adrenaline should be

administered. Treatment should be symptomatic in other cases of adverse reactions.

SIDE EFFECT

It is known that adverse reactions including mild gastrointestinal signs (diarrhoea and vomiting) and allergic reactions (skin reactions, anaphylaxis) may occur after administration of penicillins.

WITHDRAWAL TIME

Pigs (Meat and offal): 1 day

SHELF LIFE: 36 months

SHELF LIFE after first opening & reconstitution: 24 hours

STORAGE

Store in a cool dry place ($\leq 25^{\circ}\text{C}$). Strictly avoid light & heat exposure.

PACKAGING

The powder filled 50 g or 100 g in aluminum sachets, by automatic filling machine and sealing, and packed in carton box.

DATE OF REVISION: 31 May 2019

Product Registration Holder and Manufactured By:



SHENNONG ANIMAL HEALTH (M) SDN BHD
神農動物保健(馬)有限公司
No. 2, 4, 6, 8 & 10, Jalan Industri USJ 1/19,
Taman Perindustrian USJ 1,
47610 Subang Jaya, Selangor, Malaysia.